



Water Potability

In Partial Fulfillment of the Requirements in IT20L (6725)

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Presented to:

Mr. Jan Carlo Arroyo

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Rationale

A basic human need is clean and safe drinking water, yet water contamination poses a problem for communities everywhere. Contaminated water can contain dangerous bacteria, chemicals, and pollutants that, if left untreated, can cause cholera, dysentery, lead poisoning, and other illnesses. To prevent drinking tainted water that can cause health problems, it is crucial to evaluate the water's quality and determine whether it is suitable for human consumption.

In order to determine the potability of the water, this project samples the water using turbidity, pH, and chemical content measurements. Data analysis techniques can be used to identify hazardous water sources, add artificial intelligence to current approaches, and provide useful recommendations for water treatment and safety enhancements. In developing nations generally, where water contamination is a prevalent problem that both communities and authorities must deal with, this study can be very helpful in guiding decision-makers to take the appropriate steps to assess public health and improve water quality.

Data Dictionary

Field Name	Data Type	Description	Example
ph	float	pH of water, measures how acidic or alkaline (basic) the water is	8.09912418929839
Hardness	float	Capacity of water to precipitate soap in mg/L	224.236259393557
Solids	float	Total dissolved solids in ppm	19909.5417322923
Chloramines	float	Amount of Chloramines in ppm	9.27588360269408
Sulfate	float	Amount of Sulfates dissolved in mg/L	356.886135643056
Conductivity	float	Electrical conductivity of water in $\mu\text{S}/\text{cm}$	418.606213064481
Organic_carbon	float	Amount of organic carbon in ppm	16.8686369295509
Trihalomethanes	float	Amount of Trihalomethanes in $\mu\text{g}/\text{L}$	66.4200925117636
Turbidity	float	Measure of light emitting property of water in NTU	3.05593374966416
Potability	int	Indicates if water is safe for human consumption. Potable = 1 and Not potable = 0	0

Data Preprocessing and Exploratory Data Analysis

Our dataset had incomplete (missing) values in “**ph**”, “**Sulfate**”, and “**Trihalomethanes**” columns. To preprocess this, we filled it automatically with the attribute mean for all samples belonging to the same class. After that, we replaced the old values with new values in the “**Potability**” column, from “**0**” and “**1**” to “**Not potable**” and “**Potable**” respectively.

Before preprocessing:

SummaryTableChartAnalytics

3,276 Rows10 Columns

RowsColumns

Rows100Columns100

	ph # Numeric	Hardness # Numeric	Solids # Numeric	Chloramines # Numeric	Sulfate # Numeric	Conductivity # Numeric	Organic_carbon # Numeric	Trihalomethanes # Numeric	Turbidity # Numeric	Potability # Numeric
1	<NA>	204.890455471336	20,791.318980747023	7.300211873185	368.516441349803	564.308654172244	10.379783078085	86.990970461509	2.963135380632	0
2	3.716080075387	129.422920514944	18,630.057857970347	6.635245883862	<NA>	592.885359134852	15.180013116357	56.329076284518	4.500656274942	0
3	8.099124189298	224.236259393558	19,909.541732292393	9.275883602694	<NA>	418.606213064481	16.868636929551	66.420092511764	3.055933749664	0
4	8.316765884215	214.373394085623	22,018.417440775294	8.059332377439	356.886135643057	363.266516164244	18.436524495493	100.34167436508	4.628770536837	0
5	9.092223456291	181.101509236125	17,978.98633892625	6.546599974208	310.135737524204	398.410813381845	11.558279443446	31.997992727425	4.07507542543	0
6	5.584086638456	188.313323769616	28,748.68773904612	7.54486878878	326.678362911674	280.467915933488	8.399734640153	54.917861841994	2.559708227557	0
7	10.223862164529	248.07173527014	28,749.716543528233	7.513408465831	393.663395515096	283.651633507845	13.78969531752	84.603556174024	2.672988736935	0
8	8.635848718501	203.361522584571	13,672.091763901635	4.5630086856	303.309771159281	474.607644942449	12.363816698705	62.798308962925	4.401424715445	0
9	<NA>	118.988579090252	14,285.583854224516	7.804173553073	268.646940746221	389.375565871261	12.706048968658	53.928845767512	3.595017180958	0
10	11.180284470722	227.231469237975	25,484.50849098786	9.077200016914	404.04163468409	563.885481481095	17.927806411285	71.976601032219	4.370561936655	0
11	7.360640105838	165.520797259529	32,452.614409143884	7.550700906704	326.624353455602	425.383419495387	15.586810438033	78.740015664305	3.662291782852	0
12	7.974521648924	218.693304088666	18,767.65668181348	8.110384501124	<NA>	364.098230462049	14.525745697593	76.485911179652	4.01171810834	0
13	7.119824384265	156.704993340392	18,730.813653342713	3.606036090506	282.344050473961	347.715027261944	15.929535908826	79.500778336974	3.445756223322	0
14	<NA>	150.174923395136	27,331.361961927756	6.838223470688	299.415781346858	379.761834825772	19.370807181232	76.509995527958	4.413974182975	0
15	7.496232207973	205.344982158185	28,388.00488673697	5.072557773841	<NA>	444.645352332707	13.228311099225	70.300212646924	4.777382337225	0
16	6.347271760539	186.732880660576	41,065.23476453935	9.629596276481	364.48768724676	516.743281893657	11.539781191539	75.071617286638	4.376348290692	0
17	7.051785800017	211.049406060546	30,980.600786788866	10.094796011661	<NA>	315.141267244302	20.397021840722	56.651603789793	4.268428857506	0
18	9.181560007152	273.813806659801	24,041.32628006128	6.90498972647	398.350516822278	477.974641862178	13.387340780226	71.457362212952	4.503660796179	0
19	8.975464347534	279.357166770092	19,460.398131232112	6.204320858892	<NA>	431.443989990349	12.888759054304	63.821237096664	2.436085590305	0
20	7.37105030243	214.496610457157	25,630.320036999725	4.432669290372	335.754438596065	469.914551479236	12.509163940499	62.797277152661	2.560299147615	0
21	<NA>	227.435048351156	22,305.56741374141	10.333917888219	<NA>	554.820086460543	16.331693282694	45.382815177871	4.133422643579	0
22	6.660212026118	168.283746856518	30,944.363591242687	5.858769130548	310.930858317878	523.671297500944	17.884235192965	77.0423180517	3.7497012411	0
23	<NA>	215.977858688068	17,107.224225827616	5.607060453087	326.943977743867	436.256193972649	14.189062206124	59.855475826154	5.459250956029	0
24	3.902475685915	196.903246708321	21,167.500098968772	6.996311586299	<NA>	444.478882506898	16.60903315579	90.181675884745	4.528522696327	0
25	5.400301780729	140.73906225114	17,266.593421923077	10.056852484033	328.358240698684	472.874073275429	11.256381166909	56.931906445756	4.824786389768	0
26	6.514415093252	198.767351259456	21,218.702871190195	8.670936919913	323.596349010132	413.290450088535	14.899999566697	79.847842813726	5.20088507654	0
27	3.445061864385	207.926260187994	33,424.7686784948	8.782147480773	384.007005801721	441.785875673939	13.805902211271	30.284597198003	4.184396969029	0
28	<NA>	145.768180602173	13,224.935638976958	7.906444720606	304.001992797415	298.990666499932	12.729524720542	49.536848802022	4.004871127572	0

1. CSV - Local

water_potability.csv; Data Separato...
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2. Replace Values

Potability = recode(Potabiliit...

3. Create Calculation

ph = impute_na(ph, type = "...

4. Create Calculation

Sulfate = impute_na(Sulfate...

5. Create Calculation

Trihalomethanes = impute_...

+ Add a new step

After preprocessing:

Summary

Table

Chart

Analytics

3,276 Rows10 Columns

RowsColumns

RowsColumns

Rows100Columns100

	ph	Hardness	Solids	Chloramines	Sulfate	Conductivity	Organic_carbon	Trihalomethanes	Turbidity	Potability
	# Numeric	# Numeric	# Numeric	# Numeric	# Numeric	# Numeric	# Numeric	# Numeric	# Numeric	Character
1	7.080794504277	204.890455471336	20.791.318980747023	7.300211873185	368.516441349803	564.308654172244	10.379783078085	86.990970461509	2.963135380632	Not potable
2	3.716080075387	129.422920514944	18,630.057857970347	6.635245883862	333.775776610813	592.885359134852	15.180013116357	56.329076284518	4.500656274942	Not potable
3	8.099124189298	224.236259393558	19,909.541732292393	9.275883602694	333.775776610813	418.606213064481	16.868636929551	66.420092511764	3.055933749664	Not potable
4	8.316765884215	214.373394085623	22,018.417440775294	8.059332377439	356.886135643057	363.266516164244	18.436524495493	100.34167436508	4.628770536837	Not potable
5	9.092223456291	181.101509236125	17,978.98633892625	6.546599974208	310.135737524204	398.410813381845	11.558279443446	31.997992727425	4.07507542543	Not potable
6	5.584086638456	188.313323769616	28,748.68773904612	7.54486878878	326.678362911674	280.467915933488	8.399734640153	54.917861841994	2.559708227557	Not potable
7	10.223862164529	248.07173527014	28,749.716543528233	7.513408465831	393.663395515096	283.651633507845	13.78969531752	84.603556174024	2.672988736935	Not potable
8	8.635848718501	203.361522584571	13,672.091763901635	4.5630086856	303.309771159281	474.607644942449	12.363816698705	62.798308962925	4.401424715445	Not potable
9	7.080794504277	118.988579090252	14,285.583854224516	7.804173553073	268.646940746221	389.375565871261	12.706048968658	53.928845767512	3.595017180958	Not potable
10	11.180284470722	227.231469237975	25,484.50849098786	9.077200016914	404.04163468409	563.885481481095	17.927806411285	71.976601032219	4.370561936655	Not potable
11	7.360640105838	165.502797259529	32,452.614409143884	7.550700906704	326.624353455602	425.383419495387	15.586810438033	78.740015664305	3.662291782852	Not potable
12	7.974521648924	218.693300488666	18,767.65668181348	8.110384501124	333.775776610813	364.098230462049	14.525745697593	76.485911179652	4.01171810834	Not potable
13	7.119824384265	156.704993340392	18,730.813653342713	3.606036090506	282.344050473961	347.715027261944	15.929535908826	79.500778336974	3.445756223322	Not potable
14	7.080794504277	150.174923395136	27,331.361961927756	6.838223470688	299.415781346858	379.761834825772	19.370807181232	76.509995527958	4.413974182975	Not potable
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16	6.347271760539	186.732880660576	41,065.23476453935	9.629596276481	364.48768724676	516.743281893657	11.539781191539	75.071617286638	4.376348290692	Not potable
17	7.051785800017	211.049406060546	30,980.600786788866	10.094796011661	333.775776610813	315.141267244302	20.397021840722	56.651603789793	4.268428857506	Not potable
18	9.181560007152	273.813806659801	24,041.32628006128	6.90498972647	398.350516822278	477.974641862178	13.387340780226	71.457362212952	4.503660796179	Not potable
19	8.975464347534	279.357166770092	19,460.398131232112	6.204320858892	333.775776610813	431.443989990349	12.888759054304	63.821237096664	2.436085590305	Not potable
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21	7.080794504277	227.435048351156	22,305.56741374141	10.333917888219	333.775776610813	554.820086460543	16.331693282694	45.382815177871	4.133422643579	Not potable
22	6.660212026118	168.283746856518	30,944.363591242687	5.858769130548	310.930858317878	523.671297500944	17.884235192965	77.0423180517	3.7497012411	Not potable
23	7.080794504277	215.977858688068	17,107.224225827616	5.607060453087	326.943977743867	436.256193972649	14.189062206124	59.855475826154	5.459250956029	Not potable
24	3.902475685915	196.903246708321	21,167.500098968772	6.996311586299	333.775776610813	444.478882506898	16.60903315579	90.181675884745	4.528522696327	Not potable
25	5.400301780729	140.73906225114	17,266.593421923077	10.056852484033	328.358240698684	472.874073275429	11.256381166909	56.931906445756	4.824786389768	Not potable
26	6.514415093252	198.767351259456	21,218.702871190195	8.670936919913	323.596349010132	413.290450088535	14.899999566697	79.847842813726	5.20088507654	Not potable
27	3.445061864385	207.926260187994	33,424.7686784948	8.782147480773	384.007005801721	441.785875673939	13.805902211271	30.284597198003	4.184396969029	Not potable
28	7.080794504277	145.768180602173	13,224.935638976958	7.906444720606	304.001992797415	298.990666499932	12.729524720542	49.536848802022	4.004871127572	Not potable

1. CSV - Local

water_potability.csv, Data Separato...
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2. Replace Values

Potability = recode(Potabilit...

3. Create Calculation

ph = impute_na(ph, type = "...

4. Create Calculation

Sulfate = impute_na(Sulfate,...

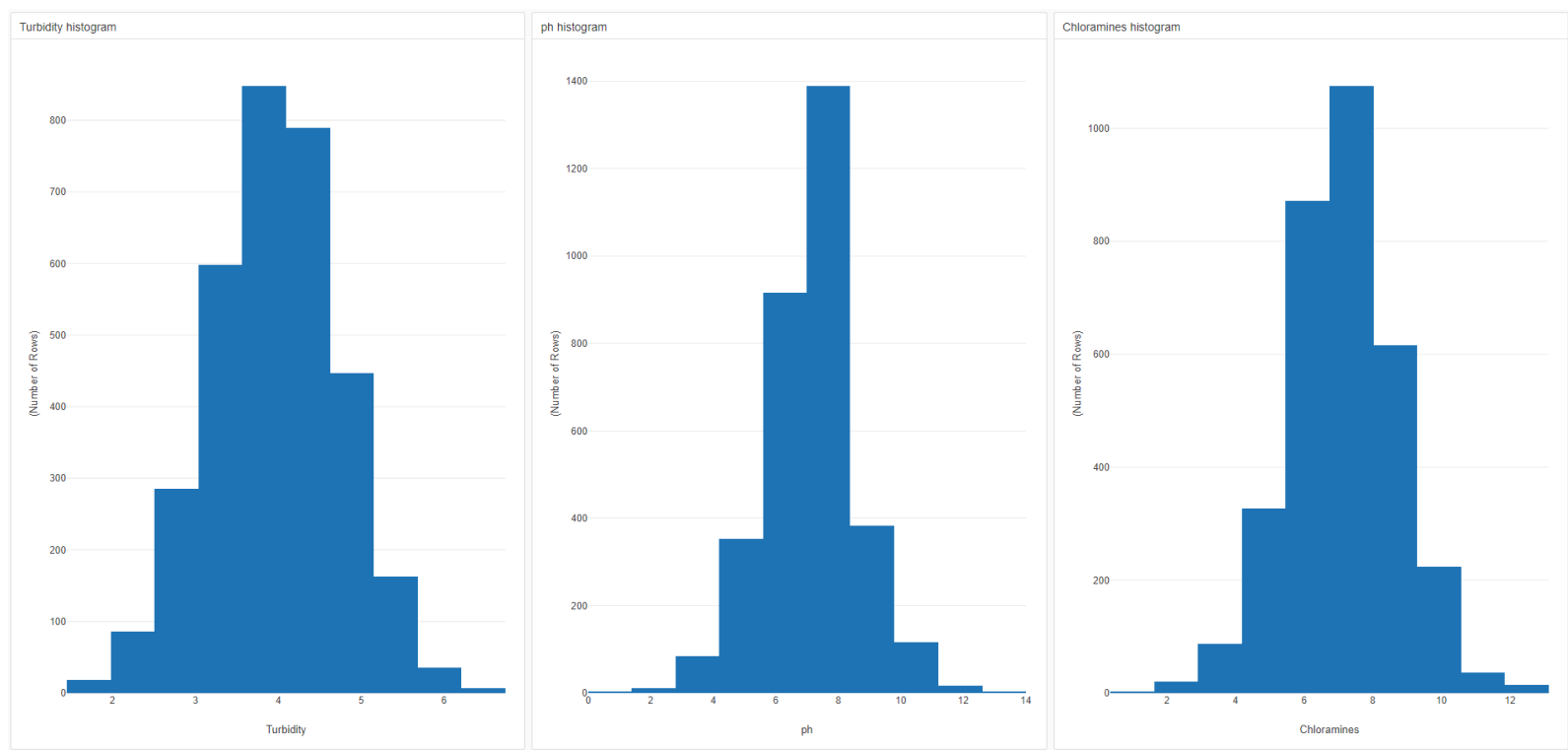
5. Create Calculation

Trihalomethanes = impute_...

+ Add New

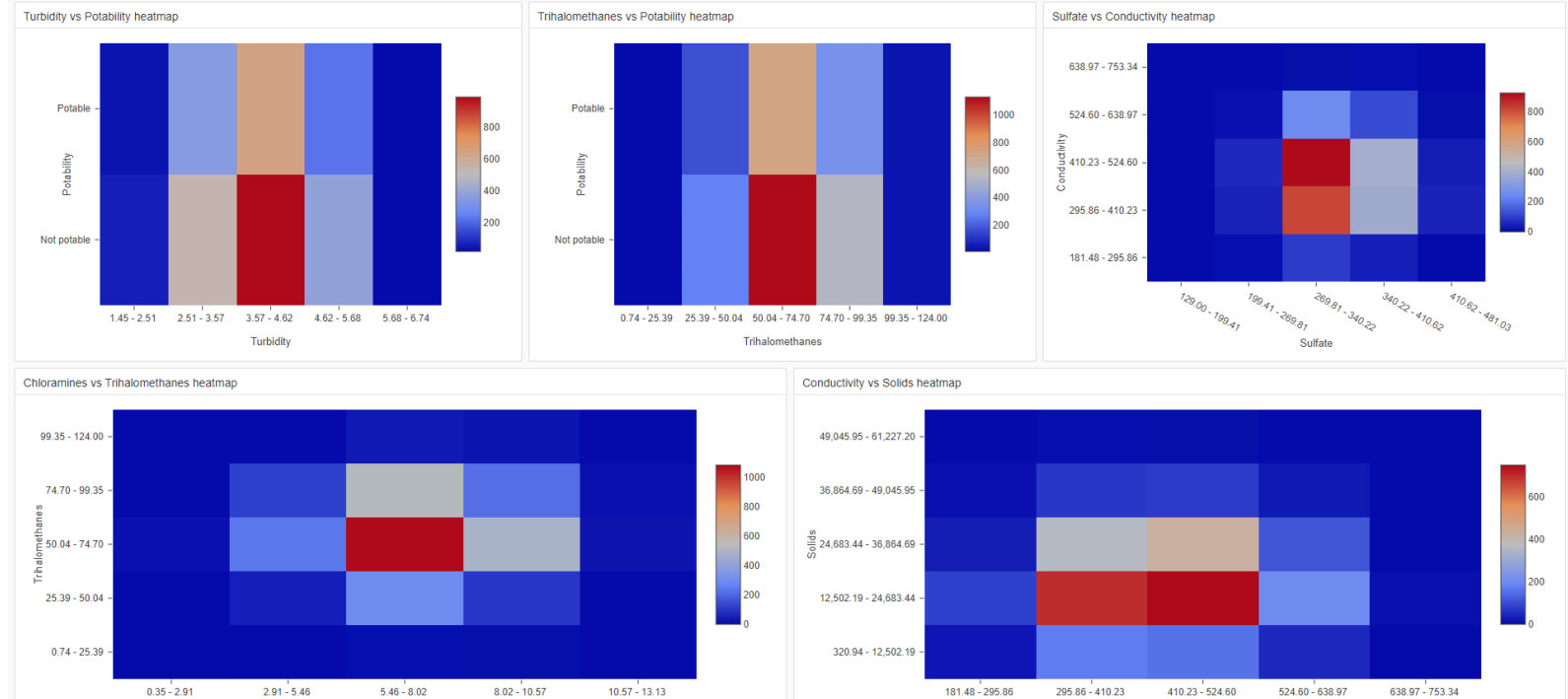
+ Add a new step

Distribution of Water Quality Parameters (Histogram)



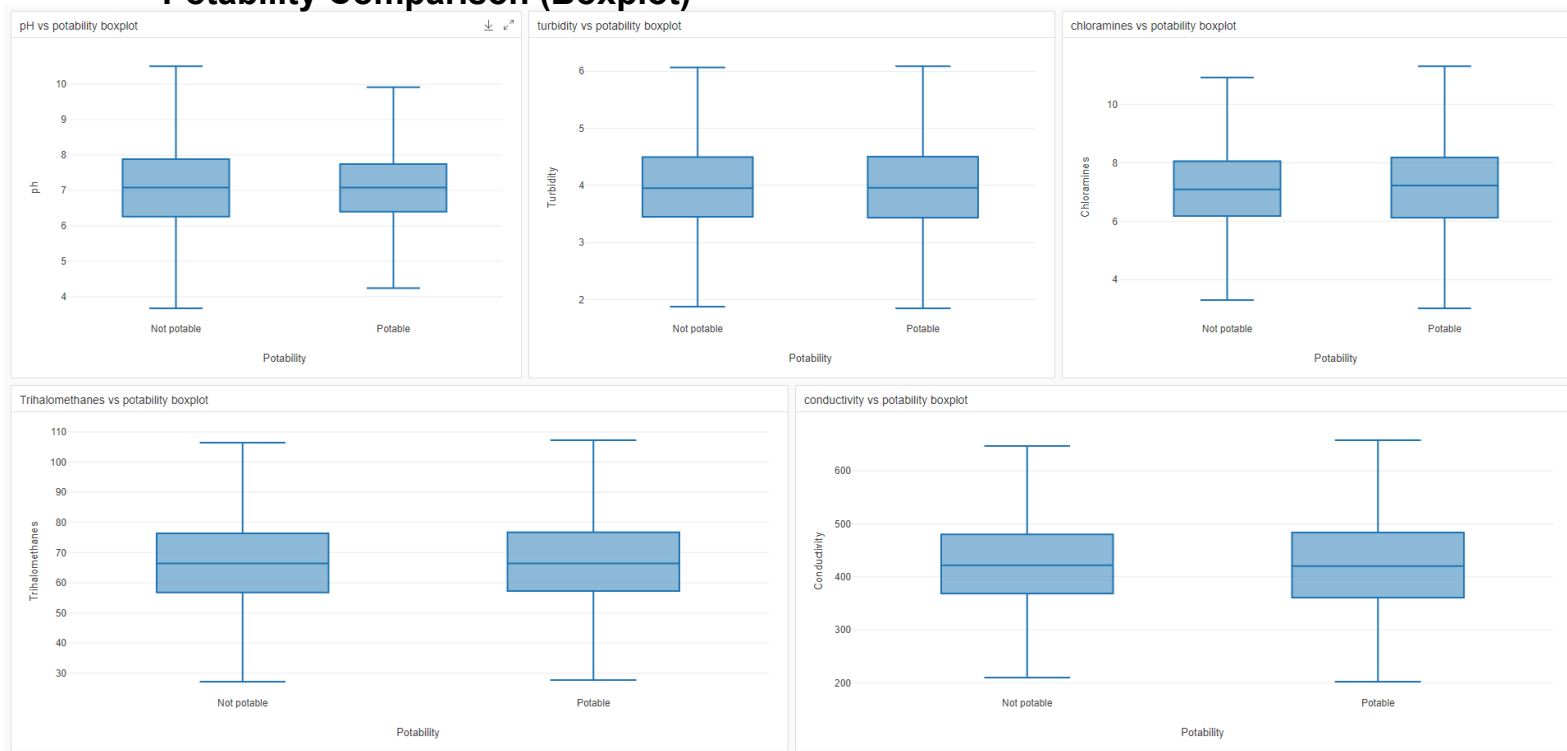
- **Turbidity:** The distribution is right-skewed, with most values between 2 and 7 NTU. Few samples have turbidity levels above 8.
- **pH:** The data is normally distributed, with most values between 6 and 9. A small number of samples have extreme pH values below 5 or above 10.
- **Chloramines:** The distribution peaks around 5-7 mg/L, with fewer samples containing very low or very high levels.

Correlation Matrix (Heatmap)



- **Turbidity vs. Potability:** Potable water samples tend to have turbidity values between 2.14 and 4.80 NTU, while non-potable water spans a wider range.
- **Trihalomethanes vs. Potability:** Non-potable samples have a higher concentration spread of trihalomethanes compared to potable samples, which tend to cluster between 20 and 60 µg/L.
- **Sulfate vs. Conductivity:** Higher sulfate concentrations (above 200 mg/L) are associated with moderate to high conductivity values, forming clusters in the middle range.
- **Chloramines vs. Trihalomethanes:** Most data points are concentrated in the 2-6 mg/L range for chloramines and 30-80 µg/L for trihalomethanes.
- **Conductivity vs. Solids:** Higher conductivity values correlate with increasing levels of dissolved solids, especially above 300 mg/L.

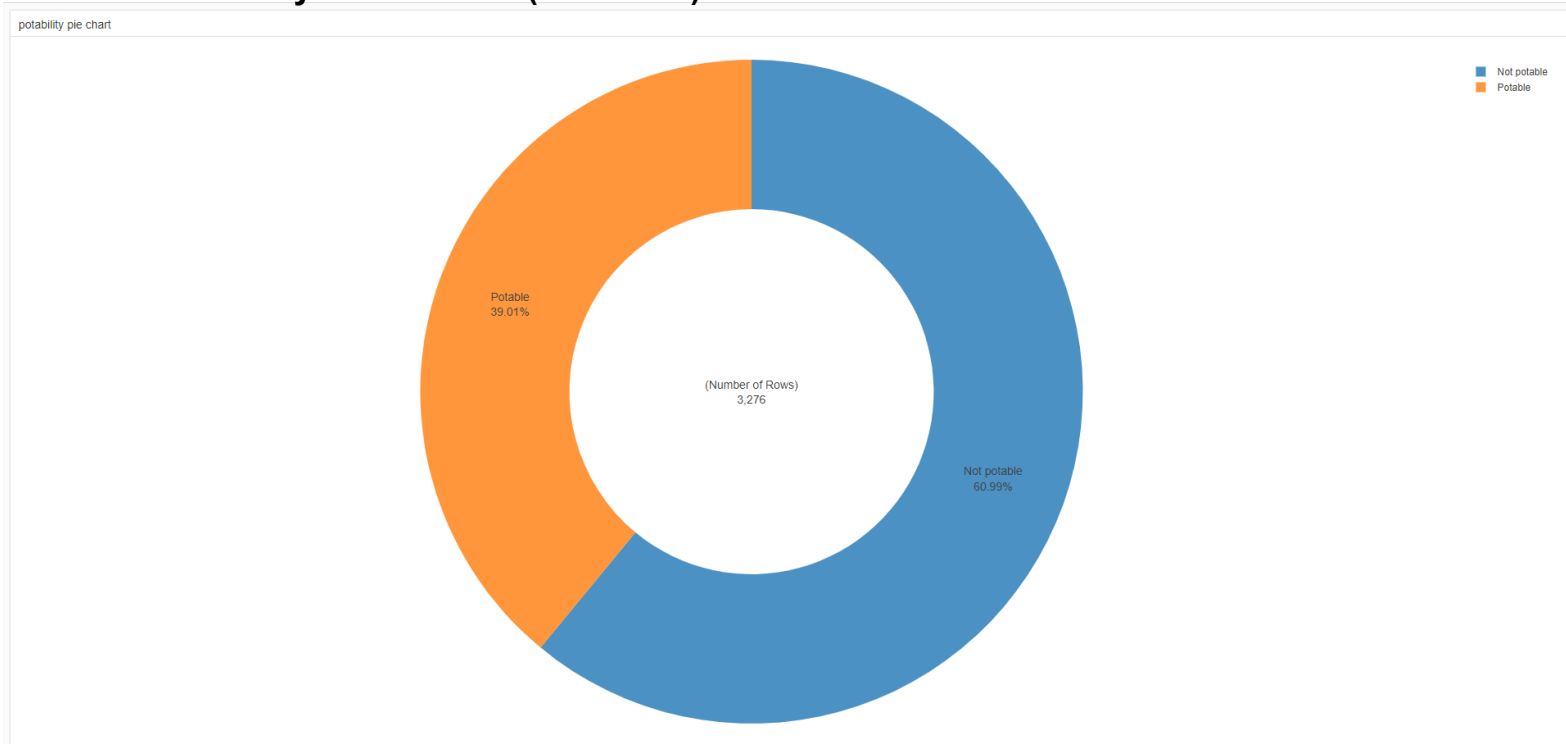
Potability Comparison (Boxplot)



- **pH vs. Potability:** The median pH is similar for potable and non-potable samples, centered around 7, but non-potable samples have a wider range.
- **Turbidity vs. Potability:** Potable water generally has lower turbidity, while non-potable water shows more variability and higher turbidity outliers.
- **Chloramines vs. Potability:** Potable water tends to have a more stable range of chloramine levels, whereas non-potable water exhibits more extreme values.

- **Trihalomethanes vs. Potability:** The median trihalomethane concentration is slightly lower in potable water, but non-potable samples show more outliers at higher concentrations.
- **Conductivity vs. Potability:** Potable water samples tend to have moderate conductivity levels, while non-potable samples show a wider range and some higher values.

Potability Distribution (Pie Chart)



60.99% of the samples are non-potable, while only 39.01% are classified as potable, indicating that a majority of the water sources in the dataset are not safe for drinking.

Python Code


```
[199] X = df.drop(columns=['Potability'])
      y = df['Potability']

      smote = SMOTE(random_state=42)
      X, y = smote.fit_resample(X, y)

      X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

      le = LabelEncoder()
      y_train = le.fit_transform(y_train)
      y_test = le.transform(y_test)

      models = {
          'Random Forest': RandomForestClassifier(n_estimators=200, max_depth=20, min_samples_split=5, min_samples_leaf=2, random_state=42),
          'XGBoost': XGBClassifier(eval_metric='logloss', n_estimators=200, max_depth=10, learning_rate=0.1),
          'LightGBM': LGBMClassifier(n_estimators=200, learning_rate=0.1, max_depth=10, force_col_wise="true", verbose = -1)
      }
```

The 3rd block of code focuses on data preparation and model initialization. It separates the dataset into predictors(X) and the target variable (y), then applies SMOTE (Synthetic Minority Over-sampling Technique) to address the potability imbalance. The dataset is split into training sets (80%) and testing sets (20%), and the target variable is encoded using LabelEncoder. Three machine learning models - Random Forest, XGBoost, and LightGBM - are initialized with specific hyperparameters to prepare for training.

```
▶ for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    recall = recall_score(y_test, y_pred, average = 'macro')
    precision = precision_score(y_test, y_pred, average = 'macro')
    f1 = f1_score(y_test, y_pred, average = 'macro')

    print(f'\n{name} Model')
    print(f"Accuracy: {accuracy * 100:.2f}%")
    print(f"Recall: {recall * 100:.2f}%")
    print(f"Precision: {precision * 100:.2f}%")
    print(f"F1: {f1 * 100:.2f}%")
```

The 4th block of code trains each of these models on the training set and evaluates their performance on the test set. It calculates and prints accuracy, recall, precision, and F1-score, providing insight into how well each model performs in predicting water potability.

```
def plot_confusion_matrix(y_true, y_pred, model_name):
    cm = confusion_matrix(y_true, y_pred)
    plt.figure(figsize=(5,4))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['Not potable', 'Potable'], yticklabels=['Not potable', 'Potable'])
    plt.xlabel('Predicted Label')
    plt.ylabel('True Label')
    plt.title(f'Confusion Matrix - {model_name}')
    plt.show()

|
for name, model in models.items():
    if name in ["Random Forest", "XGBoost", "LightGBM"]:
        y_pred = model.predict(X_test_selected)
        plot_confusion_matrix(y_test, y_pred, name)
```

The 5th block introduces a function to plot a confusion matrix, which visually represents the model's classification performance by showing the number of correct and incorrect predictions for potable and non-potable water. The function is applied to the three models, helping to analyze misclassifications.

```
single_unknown_data = {
    'ph': [7.2],
    'Hardness': [150.5],
    'Solids': [15000.5],
    'Chloramines': [4.5],
    'Sulfate': [250.3],
    'Conductivity': [450.8],
    'Organic_carbon': [12.5],
    'Trihalomethanes': [75.3],
    'Turbidity': [3.5]
}

single_unknown_df = pd.DataFrame(single_unknown_data)

for name, model in models.items():
    single_prediction = model.predict(single_unknown_df)
    print(f"Predicted Condition for the Single Data Point using {name}: ", "Potable" if single_prediction[0] == 1 else "Not Potable")
```

Finally, the 6th block demonstrates how the trained models can be used to predict the potability of a single, unseen water sample. The new data point, representing various water quality attributes, is converted into a DataFrame, and each model predicts whether the sample is potable or not potable. The results for all models are printed, allowing for a direct comparison of their predictions.

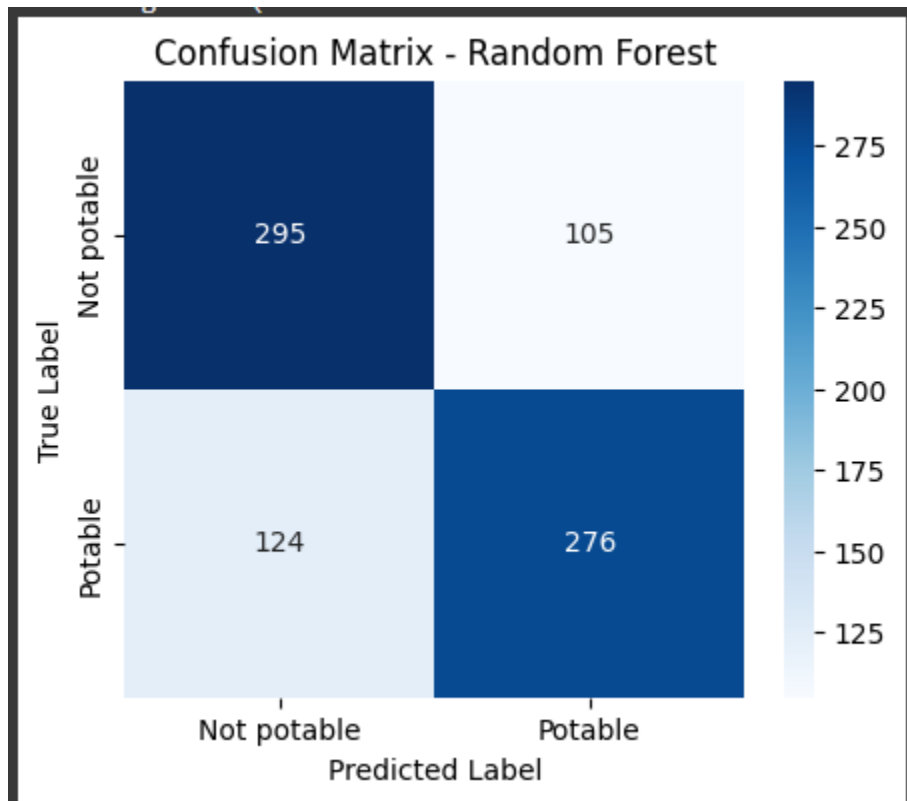
Model Performance

```
Random Forest Model
Accuracy: 71.38%
Recall: 71.38%
Precision: 71.42%
F1: 71.36%
```

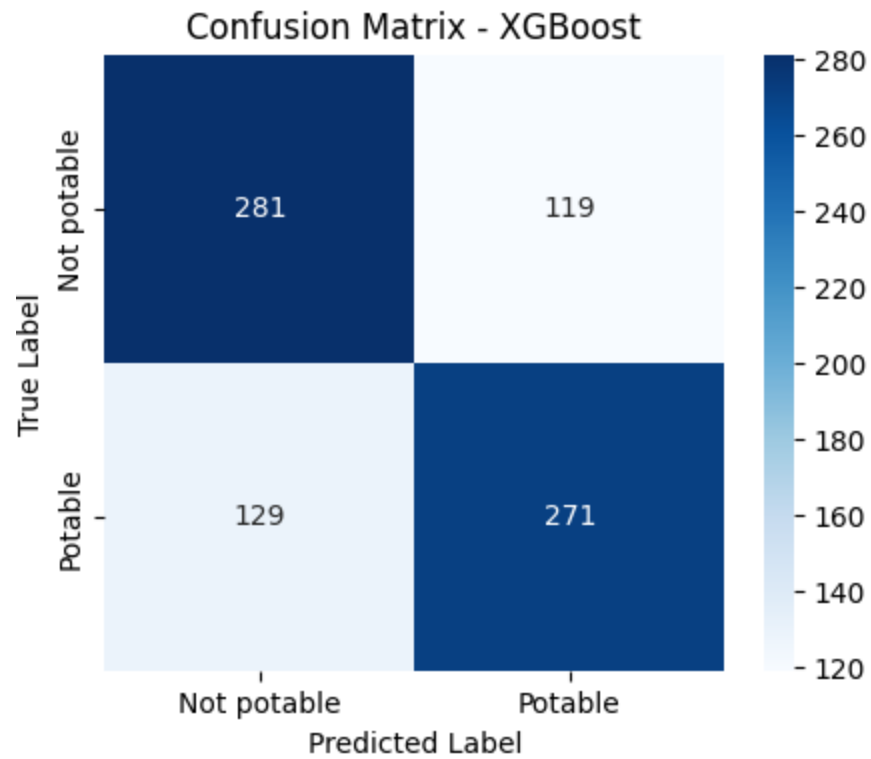
```
XGBoost Model
Accuracy: 69.00%
Recall: 69.00%
Precision: 69.01%
F1: 69.00%
```

```
LightGBM Model
Accuracy: 66.38%
Recall: 66.38%
Precision: 66.38%
F1: 66.37%
```

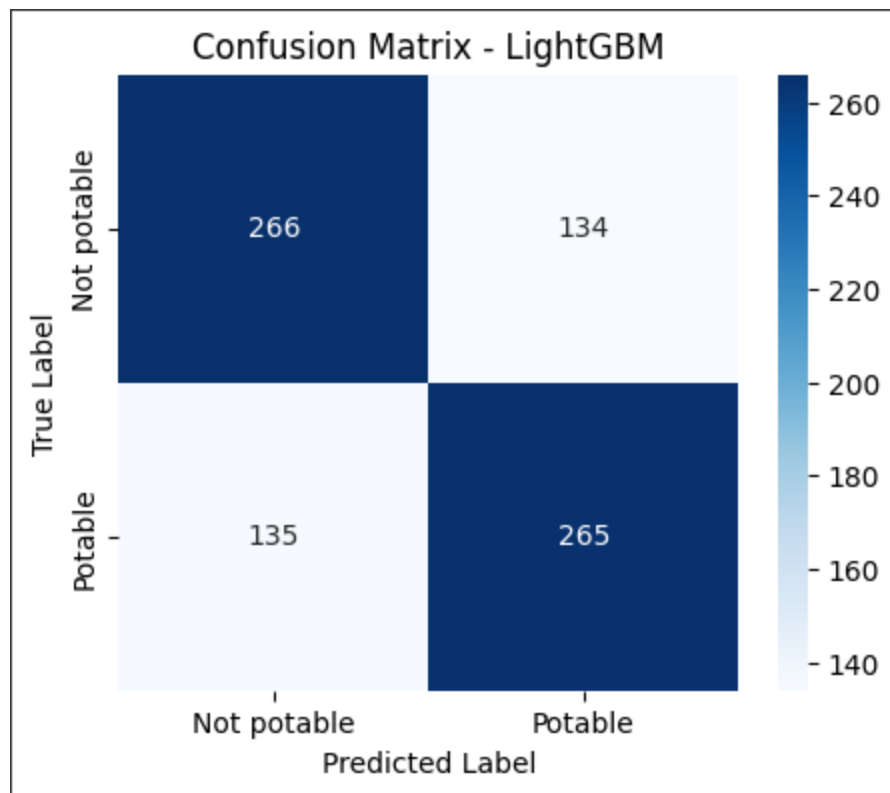
This is the accuracy, recall, precision, and f1 scores of each model in predicting water potability.



The confusion matrix for the Random Forest model shows that it correctly identified 295 non-potable water samples and 276 potable samples. However, it misclassified 105 non-potable samples as potable and 124 potable samples as non-potable. This indicates a good balance in classification but still has some errors.



For the XGBoost model, it correctly classified 281 non-potable and 271 potable water samples. However, it misclassified 119 non-potable samples as potable and 129 potable samples as non-potable. Compared to Random Forest, XGBoost performs slightly worse in detecting non-potable samples but has a similar overall accuracy.



The LightGBM model correctly identified 266 non-potable and 265 potable samples. However, it misclassified 134 non-potable samples as potable and 135 potable samples as non-potable. This model has the highest number of misclassifications among the three, suggesting that it is less reliable in distinguishing between potable and non-potable water samples.

```
single_unknown_data = {
    'ph': [7.2],
    'Hardness': [150.5],
    'Solids': [15000.5],
    'Chloramines': [4.5],
    'Sulfate': [250.3],
    'Conductivity': [450.8],
    'Organic_carbon': [12.5],
    'Trihalomethanes': [75.3],
    'Turbidity': [3.5]
}

single_unknown_df = pd.DataFrame(single_unknown_data)

for name, model in models.items():
    single_prediction = model.predict(single_unknown_df)
    print(f"Predicted Condition using {name}: ", "Potable" if single_prediction[0] == 1 else "Not Potable")

Predicted Condition using Random Forest: Potable
Predicted Condition using XGBoost: Potable
Predicted Condition using LightGBM: Potable
```

These are the predicted conditions of each model using the data we manually added on each predictors.

Discussion

Based on the specified water quality parameters, we employed three distinct machine learning models - Random Forest, XGBoost, and LightGBM - to determine whether the water is drinkable or not. Out of all the models, Random Forest performed the best when examining the confusion matrices, accurately predicting the greatest amount of potable and non-potable water. Accuracy (71.38%), precision (71.42%), recall (71.38%), and F1 score (71.36%) were its highest scores. XGBoost was slightly lower at 69%, while LightGBM had the lowest accuracy, 66.38%. Compared to the other two models, LightGBM had the highest number of false positives and false negatives, resulting in the highest misclassification as shown on the confusion matrices.

All models categorized the water as potable when the predictions for a single unknown data point were examined. However, compared to Random Forest and XGBoost, we can be less confident in LightGBM's prediction because its performance is the weakest according to evaluation metrics. The models' mistakes suggest that some non-potable water samples might have been mistakenly categorized as potable and vice versa, which could cause problems in real-world use. If there are a lot of false negatives, possibly dangerous water could be considered drinkable, which could present major health risks.

Overall, the findings show that while there is potential for improvement, the models can reasonably predict the water's potability. The prediction accuracy may be increased by adjusting the hyperparameters, choosing the best features, or using ensemble techniques. Random Forest was the best out of the 3 models and may be a dependable model based on the results thus far, but it is still unclear how well it would perform in action.

Conclusion

In order to determine whether the water is safe to drink, we evaluated three machine-learning models: Random Forest, XGBoost, and LightGBM. We checked their performance using F1-score, recall, accuracy, and precision. The Random Forest model outperformed the others with an accuracy of 71.38%, according to the results, followed by XGBoost and LightGBM with 69% and 66.38%, respectively. Although these models were successful in classifying the water samples as either potable or not, as indicated by the confusion matrices, some errors were still made.

The ability of machine learning to predict the suitability of water based on its chemical properties (such as pH, hardness, and turbidity) was one of the study's main findings. All three models were reasonably reliable because the one unknown piece of data we manually added was predicted to be drinkable. However, since misclassification still occurs in the majority of them, future research may try to improve outcomes by combining models, redefining model parameters, or adding more pertinent data.

All things considered, this project demonstrates how machine learning can help with water quality testing. More advanced iterations of these models might be applied in real-world situations to support the preservation of safe drinking water.

Links

[Google Collab Files](#)